DATA MODELLING AND VISUALIZATION LABORATORY	
Course Code: ADL48	**Credits: 0:0:1**
Pre – requisites: Basic Programming	**Contact Hours: 14P**
Course Coordinator: Dr. Sowmya B J	

Course Contents

1. Demonstrate the Python programming with: simple and complex data types, operators, flow control, string manipulation and functions
2. Implement the solution for recognizing pattern with and without using regular expressions
3. Demonstrate the operation on files: Read, Write and Organize.
4. Working with classes and familiarity with object-oriented programming concepts.
5. Working on object attributes: identity, type, and value.
6. Implementing objects and its functionalities: Iteration, object creation, object destruction, collections, attribute access, etc.
7. Demonstration of working with excel spreadsheets and web scraping
8. Demonstration of working on data set using pandas library: read_csv, describe, head, loc[:],value_counts,drop_duplicates,fillna.
9. Implement dataset‘s distribution with a histogram and Categorical Data analysis with bar plots and their ratios with pie plots using
10. Demonstrate working with line and scatter plots on a data set to explore the relationship between two attributes and visualize correlation between them.

Suggested Learning Resources

Text Books:

1. Al Sweigart, ―Automate the Boring Stuff with Python‖,1st Edition, No Starch Press, 2015. (Available under CC-BY-NC-SA license at https://automatetheboringstuff.com/)
2. Reema Thareja ―Python Programming Using Problem Solving Approach‖ Oxford University Press.
3. Allen B. Downey, ―Think Python: How to Think Like a Computer Scientist‖, 2nd Edition, Green Tea Press, 2015. (Available under CC-BY-NC license at http://greenteapress.com/thinkpython2/thinkpython2.pdf)

Course Outcomes (COs):

1. Build basic programs using fundamental programming constructs such as simple, complex data types, flow controls, regular expressions and files.(PO-1,2,3,4,5, PSO: 1,2)
2. Articulate the Object-Oriented Programming concepts such as encapsulation, inheritance and polymorphism as used in Python.(PO-1,2,3,4,5, PSO: 1,2)
3. Ability to work with a data set and perform analysis using visualization tools.(PO-1,2,3,4,5, PSO: 1,2)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment Tools	**Marks**	**Course Outcomes (COs) addressed**
Lab Test	20	CO1, CO2, CO3
Weekly Evaluation-Lab Record	30	-
The Final CIE out of 50 Marks = Marks of Lab Record + Marks scored in Lab Test		
Semester End Examination (SEE)		
Course End Examination (One full question from the Lab Question Bank, Programs will be coded using C and executed)	50	CO1, CO2, CO3